IPE Simulation — Engine Math Reference

Instructor reference (not for students)

This document maps the textbook theory each phase teaches to what the engine actually computes, and flags where the two diverge. It exists so you can answer a sharp question and never overclaim. Students never see it.

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1. Modeling philosophy

The engine is a **pedagogical simulator**, not a computable general equilibrium (CGE) or DSGE model. Every round is resolved by tractable closed-form arithmetic — Cobb-Douglas and CES aggregators, multiplicative welfare adjustments, and reduced-form policy-to-state dynamics — chosen so students can *act* and *see* a directional result, deliberately trading equilibrium rigor for legibility.

Two consequences worth holding in mind throughout:

- **Welfare is dimensionless utility**, not money. "Money" quantities (debt, taxes, side payments) reach welfare only through a stated bridge — valuing goods at `world_prices` — or are kept in a separate ledger that never touches utility.
- **Prices are not market-clearing.** Inter-country trade is barter at student-negotiated ratios; there is no endogenous price vector and no factor-price equalization. The engine demonstrates each model's qualitative lesson; it does not solve for equilibrium.

Notation

g indexes goods (cloth, wine, machinery); j indexes the J goods in welfare; L, K are labor and capital; Y_g is output of good g; c_j is consumption of good j; w, r are wage and return to capital; D is a country's debt stock; d is its currency depreciation factor (1.00 = par).

Parameters (all tunable at the top of `engine.py`)

Symbol / constant	Value	Meaning
factor shares (cloth)	α=0.70, β=0.30	labor-intensive
factor shares (wine)	α=0.55, β=0.45	moderate
factor shares (machinery)	α=0.25, β=0.75	capital-intensive
`WORLD_PRICES`	cloth 1, wine 1, machinery 1.5	numeraire goods prices
productivity tiers	HIGH 1.3, MED 1.0, LOW 0.7	firm productivity (Ph. 4)
`VARIETY_RHO` ρ	0.75	CES; $\sigma = 1/(1-\rho) = 4$
`FIRM_LOCAL_SHARE` λ	0.2	MNC output sold locally
`PHASE5_MONEY_GROWTH_CHOICES`	0, 2, 5, 10%	money-supply growth
`WARNING_DEVALUATION`	0.90	stress-1 currency drop
`CRISIS_DEVALUATION`	0.70	stress-2 currency drop
`CRISIS_WELFARE_HIT`	0.10	stress-2 welfare hit
`BASE_FX_FRICTION`	0.02	cross-currency friction
`WARNING_FRICTION_BUMP`	0.03	friction after a warning
`DEBT_BASE_RATE`	0.05	base sovereign rate
`DEBT_RISK_PREMIUM`	0.15	premium × (debt/capacity)
`DEBT_DEFAULT_BAN_ROUNDS`	2	borrowing ban after default
`DEBT_DEFAULT_FRICTION`	0.05	defaulter trade friction
`DEBT_DIVIDEND_PENALTY`	0.5	WTO dividend halved in ban
`IMF_DEBT_RELIEF`	0.5	bailout refinances half the stock
`IMF_AUSTERITY`	0.08	austerity welfare cut
`IMF_AUSTERITY_ROUNDS`	2	austerity duration
`WTO_DIVIDEND`	0.02	member-member friction cut
`HEGEMON_PROVISION_COST`	0.04	hegemon's welfare cost to lead
`HEGEMON_PROVISION_BENEFIT`	0.05	global friction cut if providing
`HEGEMON_WITHHOLD_PENALTY`	0.07	global friction rise if withholding

The welfare pipeline (order of operations each round)

For each country the engine computes base utility, then applies multiplicative adjustments in this fixed order (later phases only):

$$U_{\text{base}} \xrightarrow{\text{Ph.5 crisis}} \xrightarrow{\text{Ph.7 hegemon cost}} \xrightarrow{\text{Ph.7 global crisis}} \underbrace{U_{\text{trade}}}_{\text{gains-from-trade measured here}} \xrightarrow{\text{Ph.6 debt}} U_{\text{final}}.$$

Gains-from-trade are measured at U_{trade} — before debt — so that borrowing (consumption pulled from the future) is not miscounted as a gain from trade.

2. Phase 1 — Ricardian comparative advantage

Theory. One factor (labor), constant productivity per good. A country exports the good in which it has lower opportunity cost. The PPF is a straight line; gains from trade come from specialization.

Engine (`_compute_production`, `_utility`).

$$Y_g = L_g \cdot a_g \qquad \text{(linear: } a_g = \text{productivity of good } g\text{)}$$

$$U = \prod_{j=1}^{J} c_j^{1/J} \qquad \text{(Cobb-Douglas, equal weights)}$$

Opportunity cost (shown in the briefs, 2-good case): OC $= a_{g_0}/a_{g_1}$. Trade is barter; a tariff destroys a fraction of the imported good, received $=$ qty $(1-t)$.

Divergence. Linear production ⇒ constant opportunity cost (no diminishing returns, no curved PPF). "Gains from trade" is utility of the consumed bundle minus utility of own production (the no-trade baseline), not a welfare-theoretic equivalent variation.

3. Phase 2 — Heckscher-Ohlin

Theory. Two factors. A country exports the good that intensively uses its abundant factor. Stolper-Samuelson: opening trade raises the real return of the abundant factor and lowers the scarce factor (the magnification effect: $\hat{p}_1 \Rightarrow \hat{w} > \hat{p}_1 > 0 > \hat{r}$ when good 1 is labor-intensive).

Engine (`_compute_production`, `_shadow_prices`, `_compute_factor_prices`). Production is Cobb-Douglas; factor returns are *value* marginal products at implicit home prices taken from the post-trade consumption bundle c:

$$
\begin{aligned}
&Y_g = \mathrm{TFP}_g \cdot L_g^{\alpha_g} K_g^{\beta_g}, \qquad \alpha_g + \beta_g = 1 \\
&p_g = \min\!\left(\frac{U}{c_g},\ 10\right), \qquad U = \Big(\prod_j c_j\Big)^{1/J} \qquad (\text{so } \textstyle\prod_j p_j^{1/J} = 1) \\
&w_g = p_g\,\alpha_g \frac{Y_g}{L_g}, \qquad r_g = p_g\,\beta_g \frac{Y_g}{K_g}
\end{aligned}
$$

The reported w and r are allocation-weighted averages of w_g and r_g. p_g is the Cobb-Douglas marginal utility $U/(Jc_g)$ rescaled so the ideal price index is 1: a good the country consumes little of is dear at home, and returns are measured in consumption bundles — *real* returns. In autarky $c = Y$, so prices reflect domestic scarcity. Opening trade changes c, hence p, hence w and r: exporting the labor-intensive good raises its home price and with it the wage. *That is the Stolper-Samuelson channel.* The cap of 10 binds only when a good is exported almost entirely.

Why this form. The engine originally reported *physical* marginal products, $w_g = \alpha_g Y_g/L_g$. Those depend only on the allocation, so trade could never move them — in class a country that traded with an unchanged allocation showed identical factor prices to three decimals — and averaging cloth-units with machinery-units is not meaningful. A first fix scaled prices so the bundle was worth its world-price value; nominal returns then swelled and shrank with the gains from trade, and a capital-abundant country could show its return to capital *falling* when trade opened. Normalizing the price index to 1 fixes that: in a controlled test with every allocation held fixed, the labor-abundant countries' wages rise and capital returns fall, and the capital-abundant country's capital return rises and wage falls, in absolute real terms (`stress_test.py`, `test_stolper_samuelson`). Histories saved before this change keep the old numbers; replaying those rounds gives the new ones.

Divergence. The prices are shadow prices read off a consumption bundle, not market-clearing prices: there is no zero-profit condition, and because students allocate factors freely, sector marginal products are not equalized — the reported w and r are averages across sectors, not a single economy-wide factor price. No factor-price equalization. The magnification effect holds in the controlled test but is not guaranteed for arbitrary play: when a country reallocates its factors *and* trades in the same round, the reallocation effect and the trade effect are mixed together. To show Stolper-Samuelson cleanly, compare a country whose allocation did not change. Trade is also barter at whatever terms students negotiate, not balanced at common prices, so a lopsided deal is a transfer: w and r value what the country *produces* at home prices, including goods shipped out for little in

return, and both can rise while welfare falls. (In class, a country that sent 27 cloth for 9 machinery saw its wage double while its welfare fell 23%.) Welfare, not w and r, says whether a deal paid. The engine will not reproduce, e.g., the Leontief paradox.

4. Phase 3 — MNCs, varieties, new trade theory

Theory. Krugman: increasing returns + love of variety yield intra-industry trade and gains from variety, formalized with CES preferences and monopolistic competition.

Engine (`_compute_firm_production`, `_build_variety_bundles`, `_utility_with_varieties`, `_compute_firm_profits`). MNC output is linear in scale, $Y_{\text{firm}} = \text{scale} \times \text{productivity}$ (0 in a relocation round; scale capped at `max_scale`). Only the firm's local sales, λY_{firm} with $\lambda = 0.2$ (`FIRM_LOCAL_SHARE`), enter the host's production and consumption; the rest sells on world markets, and the owner's revenue counts all of it. Each good's consumption is a CES aggregate over its varieties, nested inside the Cobb-Douglas:

$$c_g = \Big(\sum_v q_v^{\rho}\Big)^{1/\rho}, \qquad U = \prod_g c_g^{1/J}, \qquad \rho = 0.75, \quad \sigma = \tfrac{1}{1-\rho} = 4.$$

More varieties raise c_g at fixed total quantity — the variety gain. The no-trade baseline is the country's *own* output, without its hosted firms' local sales: foreign capital producing on its soil is part of being open, so a country that attracts a firm gains and one that loses a firm loses. Firm profit (numeraire): $\pi = Y_{\text{firm}} \cdot p_{\text{industry}} - \text{scale} \cdot \text{unit_cost}$. Trade moves a good's varieties in proportion to the exporter's variety mix.

MNC tax (Phase 3+). Each host h sets a rate $t_h \in [0, 0.5]$ on foreign-owned firms' revenue R (premium included in Phase 4+); a populist government imposes a floor, so the rate in force is $\tau_h = \max(t_h, \underline{t}_h)$. The owner pays $\tau_h R$; the host collects $T_h = \sum_{f \in h} \tau_h R_f$ and keeps it:

$$U_h \leftarrow U_h\Big(1 + \frac{T_h}{C_h}\Big), \qquad C_h = \sum_g c_{h,g}\, p_g,$$

applied after the gains-from-trade metric (a transfer from foreign owners, not a gain from trade). The brake is mobility: a firm that relocates takes its output, and so its variety, with it, which costs the host far more than the tax brings in. A relocation to the owner's home country is refused (no re-shoring).

Divergence. Real new trade theory has monopolistic competition with free entry, markup pricing, and an *endogenous* number of varieties pinned down by market size and fixed costs. Here the firm set is fixed and CES enters only on the consumption side; firms sell at a fixed world price, not a markup. The model captures love-of-variety and intra-industry trade qualitatively, not the monopolistic-competition equilibrium.

5. Phase 4 — Heterogeneous firms (Melitz)

Theory. Firms draw productivity φ; only firms above a cutoff φ^* find it worth paying the fixed export cost f_x and exporting. Trade reallocates toward high-φ firms and raises average productivity (selection).

Engine (`_compute_firm_profits`). Each firm has a fixed productivity tier φ and makes $Y = \text{scale} \cdot \varphi$ units. With `export` = True it reaches foreign buyers, so every unit sells at a premium m, and it

pays the fixed cost:

$$\pi = \underbrace{Y\, p_{\text{ind}}\,(1+me)}_{\text{revenue}} - \text{scale} \cdot \text{unit_cost} - \underbrace{f_x e}_{\text{if exporting}} - \underbrace{\text{revenue} \cdot \tau_{\text{mnc}}}_{\text{MNC tax}}, \qquad m = \bar{m}\left(1 - \bar{\tau}_{h,g}\right),$$

where $e = 1$ if the firm exports and 0 otherwise, with $\bar{m} = 0.25$ (`EXPORT_MARKET_GAIN`) and $\bar{\tau}_{h,g}$ the mean tariff the other countries applied *last round* to good g from the firm's host h (declared bilateral rate, raised to any populist floor). Exporting pays iff $m\,Y\,p_{\text{ind}} > f_x$. The shipped roster sets $f_x/p_{\text{ind}} = 8$ in every industry, so at $\bar{\tau} = 0$ the cutoff is $Y > 32$: at full scale (40) HIGH firms ship 52 units and MED 40, both above it; LOW ship 28 and cannot reach it at any scale. At $\bar{\tau} = 0.2$, MED firms break exactly even; at $\bar{\tau} = 0.5$ ($m = 0.125$, cutoff 64) no firm's exports pay. Populist regimes set a tariff floor (applied tariff = max(declared, floor)) and a minimum MNC tax (Phase 3); the tax falls on premium-inclusive revenue.

Why this form. A fixed cost with no upside makes exporting a dominated choice, and selection never happens. A revenue premium is the smallest change that creates a genuine cutoff; tying it to the tariff wall links Phase 2 trade politics to Phase 4 firm decisions — protection abroad removes the MED exporters first. Using last round's tariffs makes m a number on the board, so the decision is computable at the desk.

Divergence. Real Melitz derives φ^* endogenously from the productivity distribution and fixed costs in equilibrium, and exporters sell more units, not the same units at a higher price. Here productivity is discrete (HIGH/MED/LOW), the premium is a profit-ledger effect only (host welfare counts the firm's units whether or not it exports), and the export choice is the student's, so selection is shown, not solved.

The reserve currency (`award_reserve_currency`, end of Phase 4) goes to the best average gain from trade over all rounds. New games (`"best_mix"`) measure each round against the country's best mix with no trade, $\bar{U}_i$: the split of its own factors that maximises $\prod_g Y_g^{1/J}$ (`best_autarky`). Taking logs, $\sum_g \frac{1}{J}(\ln A_g + \alpha_g \ln L_g + \beta_g \ln K_g)$ separates by factor, so

$$L_g = L\,\frac{\alpha_g}{\sum_h \alpha_h}, \qquad K_g = K\,\frac{\beta_g}{\sum_h \beta_h}$$

(an equal split of the workers in Phase 1). Round t scores $\left(W_{it}/\bar{U}_{it} - 1\right)\sqrt{\bar{U}_{it}/\bar{\bar{U}}_t}$, the root of the country's size relative to the average. Measured against what a country chose to produce instead, hard specialisation lowers the bar it is judged against and a small economy gains more in percent; the older rules (`"gains"`, the plain average against own output, and `"half_size"`) remain available, and a game saved before the rule existed keeps `"gains"`.

6. Political approval (all phases)

Theory. In welfare terms a tariff is pure deadweight loss, so a welfare-maximizing player would never protect — which makes the politics of Blocks 4, 6 and 7 unplayable. The missing half is the *distributional* one: protection is a transfer to the import-competing sector, whose gain is concentrated and organized while the consumer's loss is diffuse (Olson). This ledger supplies that political return without touching the utility function.

Engine (`_update_approval`, `_resolve_governments`). For country i, with M_i the value of imports actually received, $\bar{t}_i$ the import-weighted mean tariff levied on them, C_i consumption capacity at

world prices, c_i the share of consumption paid to the groups trade displaced, S_i the *net* goods another country sent (received minus sent), and A_i approval:

$$\text{exposure } x_i = M_i/C_i, \qquad \bar{t}_i = \text{(tariff-weighted)}$$

$$\Delta A_i = \underbrace{\pi\,\bar{t}_i x_i}_{\text{protection}} - \underbrace{\varepsilon\,x_i}_{\text{openness}} + \underbrace{\text{clip}\Big(\rho\tfrac{U_i - U_i'}{U_i'}\Big)}_{\text{prosperity}} + \underbrace{\kappa\,(c_i + S_i/C_i)}_{\text{compensation}} + \underbrace{\delta\,(A_0 - A_i)}_{\text{drift}}$$

$$A_i \leftarrow \text{clip}\,(A_i + \Delta A_i,\ 0,\ 100)$$

with $\pi = 35$, $\varepsilon = 14$, $\rho = 35$ (clipped to ± 15), $\kappa = 25$, $\delta = 0.10$, $A_0 = 50$, and U_i' last round's welfare. The prosperity term is zero in the first round of a phase: each phase measures welfare differently, so a jump across the boundary is the model changing, not the country prospering.

Buying off the losers. $c_i \le 0.25$ is a decision, taken on the same form as tariffs and available in every phase. It costs welfare — $U_i \times= (1 - 0.20\,c_i)$, the deadweight on taxing and transferring — but leaves trade open, so at a typical import exposure it buys about the approval a 20% tariff does for roughly two-thirds of the welfare, and without shrinking anyone's trade. That is the "compensate rather than protect" move of Block 7 made playable. Side payments (S_i) are goods one country hands another, also available from Phase 1 on; only the *net* counts, so two countries cannot pass the same crate back and forth and both bank approval for it, and only payments actually made count (one the donor cannot cover fails and buys nothing). Side payments that go both ways between two countries in one round are a trade in disguise, and are settled as one: a good sent both ways nets out, and the rest pays tariffs, FX friction and currency costs and counts in M_i, not S_i (several goods split into one trade per pair of goods, by world-price value).

The endogenous backlash. If a round ends with $A_i < 30$ the government falls (one round is enough; `APPROVAL_CRISIS_ROUNDS`): `inject_populist_backlash` fires automatically (20% tariff floor, 10% MNC tax) and A_i resets to A_0. Protection is therefore imposed *on* a country that ignores its losers — the instructor does not choose the victim. The populist government governs for two rounds (`POPULIST_ROUNDS`), then steps down and the floor and the MNC tax minimum lift: one round leaves no time to answer it, three is most of a phase.

Tuning. Holding policy fixed, approval converges to

$$A^* = A_0 - \frac{x\,(\varepsilon - \pi\bar{t})}{\delta},$$

so the steady state, not the starting value, decides who survives. The coefficients are set so A^* sits below the crisis floor under free trade (≈ 20) and above it from roughly a 20% tariff (≈ 35). Under uniform play over a 20-round term this yields: free trade highest welfare but one government falls; 20% tariffs all survive at $\sim 3\%$ less welfare; 40% tariffs politically safe at $\sim 12\%$ less. *An earlier tuning put A^* below the floor at every tariff level, so every trading country fell regardless of policy — a doom clock rather than a decision. Check A^*, not the first few rounds, after changing any coefficient.*

Divergence. This is the most openly *reduced-form* component in the engine — a stipulated response function, in the same family as trilemma stress and the debt risk premium, not something derived from preferences or voting. There is no median voter, no electoral rule, and no separate interest groups: the "concentrated benefit" is a coefficient, not an aggregation over agents. It also has no free-rider problem and no lobbying cost, so it dramatizes Olson's logic rather than modelling it. Approval never enters U, so all welfare and gains-from-trade figures remain exactly as documented above; it is a second scoreboard, deliberately in tension with the first.

7. Phase 5 — Money, FX, and the trilemma

Theory. The Mundell-Fleming "impossible trinity": a country cannot simultaneously run a fixed exchange rate, open capital markets, and independent monetary policy. Pushing all three invites a speculative attack.

Engine (`_apply_monetary_decisions`, `_compute_fx_friction`, `_weak_fx_cost`). Each round a country picks a regime (peg or float), capital controls (on or off) and money growth $g \in \{-2, 0, 2, 5, 10\%\}$. Printing ($g > 0$) or tightening ($g < 0$) *is* independent monetary policy; $g = 0$ follows the anchor. Only printing can overreach: tight money invites no run on the currency. Let s be the stress counter and d the FX index (depreciation factor, 1 = par). Overreach is peg $\wedge$ open capital $\wedge\ g > 0$. Per round:

$$\text{overreach} \Rightarrow s \mathrel{+}= 1; \qquad \text{else } s = 0.$$

$$s = 1:\ d \times= 0.90 \text{ (warning)}; \qquad s = 2:\ d \times= 0.70,\ U \times= 0.90,\ s \to 0 \text{ (crisis)}.$$

What each choice buys or costs:

$$\begin{aligned}
\text{printing } g > 0: &\quad d \times= (1-g), \qquad U \times= (1+\sigma\, g\, d), \quad \sigma = 1 \\
\text{tightening } g \in \{-0.02, -0.05, -0.10\}: &\quad d \times= (1-g) \text{ (it can pass par)}, \qquad U \times= (1+\sigma\, g) \\
\text{weak currency:} &\quad \text{each import received} \times \big(1 - \lambda\,(1 - d_{\text{receiver}})\big), \quad \lambda = 0.5 \\
&\quad U \times= \big(1 - \mu\,(1-d)\big) \text{ every round}, \quad \mu = 0.10 \\
&\quad U \times= \big(1 + \varepsilon\,(1-d)\,x\big) \text{ every round}, \quad \varepsilon = 0.2 \\
\text{capital controls:} &\quad \text{hosted firms' output} \times (1 - 0.15), \text{ no foreign borrowing} \\
\text{FX friction:} &\quad \text{fx} = 0.02 \ (+0.03 \text{ just after a warning}) \\
\text{a peg:} &\quad d \times= d^R_t / d^R_{t-1}
\end{aligned}$$

A peg is fixed to the reserve currency R, so it imports the anchor's monetary policy: after its own moves, a pegged currency falls by whatever the reserve currency lost since the peg last tracked it — the anchor's printing, a warning or crisis, an attack between rounds. Tracking starts the round a country pegs (no retroactive drop) and stops when it floats; the anchor's own union does not follow itself. FX friction is zero with the reserve holder, inside a monetary union, or between two pegged currencies. Friction and the weak-currency cut stack on tariffs multiplicatively: $\text{received} = \text{qty}\,(1-t)(1-\text{fx})\big(1-\lambda(1-d)\big)$. In the export gain, x is the share of the country's own output (at world prices) that it shipped this round in trades with countries outside its monetary union (`_export_share`); side payments are gifts, not exports. A weak currency makes exports cheap abroad, so it now pays a country that exports while it costs one that imports. The stimulus, the drag and the export gain are applied after the gains-from-trade measure (none is a gain from trade); the drag hits any currency below par, printer or crisis victim. Crisis targeting (`fx_vulnerability`) scores overreach, stress, a recent warning, an open peg, loose money, open capital and a weak currency; a dead heat at the top, or no exposure at all, fires nothing. `"managed"` is accepted as a synonym for `"float"`.

Challenging for the reserve currency (`_settle_reserve_challenge`). Each round a country may name the country whose challenge it backs, or itself to challenge (the holder's own challenge is void). After the round, a challenger wins if its coalition C, itself and its backers, passes the game's rule (`challenge_rule`) among the N countries:

$$\texttt{"majority"} \text{ (default): } |C| > N/2, \qquad \texttt{"welfare"}: \ \sum_{i \in C} \sum_{s \le t} W_{is} > \sum_{i \notin C} \sum_{s \le t} W_{is},$$

with cumulative welfare counted through this round. The winner is R from the next round. Two coalitions cannot both pass either test, so at most one challenge wins.

Why this form. Under the original rules printing only depreciated the currency, independence only enabled printing, capital controls cost nothing, and `managed` behaved exactly like `float` — so peg + controls + 0% was never worse than any other choice, and the trilemma never bit. Each option now buys something. The stimulus is scaled by d because a flat stimulus made printing 10% every round the best strategy even as the currency sank to a quarter of par; the import cost falls on the printer itself, because a cost borne only by foreign firm owners would not deter it; and the welfare drag exists because, in a world with little or no trade, the import cost barely bites and printing forever won again. The export gain exists because a weak currency was otherwise all cost, and the larger tightening steps because climbing back from a crisis at 2% a round took most of a term. `stress_test_money_balance.py` plays six quite different worlds (class-like, heavy, light and no trade, specialized, six countries) and fails if printing forever or peg + controls + 0% is ever a country's best choice, if one choice is best for everybody in a world, or if controls fall more than 12% behind. The constants sit in the middle of the passing range (it holds with σ and μ both scaled by $\pm 25\%$ and with λ from 0.35 to 0.65). The export gain passes from 0 to 0.3. At 0.35 printing 10% every round starts to pay for a big exporter, and 0.25 fails once λ falls to 0.35, so $\varepsilon = 0.2$ holds under all of those changes. A real class's trade pattern should not need a re-tune. In that test, printing every round pays only for economies that barely trade, or by a hair (under 1%) for a small economy that exports much of its output; in trading worlds the best fixed stance is otherwise a peg or a float at 0%, depending on partners — the small-open-economy lesson — and printing is a one-off tool.

Divergence. d is a *reduced-form currency-strength index* driven by policy choices and shocks — there is no money supply/demand, no interest parity, no PPP, no modeled exchange-rate market. The stimulus is a reduced-form short-run gain from monetary expansion (no Phillips curve, no output gap); scaling it by d stands in for lost credibility. The trilemma is operationalized as a stress accumulator, not derived from an open-economy equilibrium. A monetary union shares one (s, d, policy) state across members (individual debts, shared currency — the Eurozone analogy).

8. Phase 6 — Sovereign debt

Theory. Debt is sustainable while the interest burden stays within capacity; spreads rise with the debt burden; "original sin" (Eichengreen-Hausmann) means foreign-currency debt becomes crushing when the local currency depreciates.

Engine (`_apply_country_debt`). Let consumption capacity $C = \sum_g c_g\, p_g$ (goods at world prices). Then:

$$\text{rate} = \underbrace{0.05}_{\text{base}} + \underbrace{0.15}_{\text{premium}} \cdot \frac{D}{C}, \qquad \text{interest} = D \cdot \text{rate}$$

$$\text{borrow } B: \quad U \times= \left(1 + \tfrac{B}{C}\right), \quad B \le 0.25\, C, \quad B = 0 \text{ under capital controls}$$

$$\text{service (interest + repay)}: \quad U \times= \max\left(0,\ 1 - \tfrac{\text{service}/\hat{d}}{C}\right), \qquad \hat{d} = d/d^R, \qquad D \leftarrow D + B - \text{repay}$$

$$\text{default}: \quad U \times= 1 - \min\left(1, 1.25\, \tfrac{D}{C}\right), \qquad D \to 0$$

The division of service by $\hat{d}$ *is* original sin. The debt is owed in the reserve currency R, so what counts is a currency's value against it, $\hat{d} = d/d^R$ (`_fx_vs_reserve`): a currency that falls against

the reserve ($\hat{d} < 1$) inflates the real burden of the debt. The reserve holder owes in its own money ($\hat{d} = 1$); a peg that falls with the reserve owes no more; a currency that holds while the reserve falls ($\hat{d} > 1$) owes less. With a calm reserve ($d^R = 1$) this is the original rule, service/d. **Default** costs 1.25 times the debt's face value (*not* divided by d), bans borrowing for 2 rounds and adds trade friction. At par it therefore costs more than repaying; after a large devaluation ($1/d$ well above 1.25, a fall of roughly 20% or more) it escapes the magnification — which is when walking away pays. In `final_round` there is no new borrowing and the whole stock is repaid (or defaulted on). **IMF bailout**: $D \times= 0.5$ now, in exchange for $U \times= 0.92$ for 2 rounds (austerity).

Why this form. Borrowing used to be capped at C (enough to double a round's welfare) and default cost only the ban and the friction: in a six-round test, a country that borrowed the maximum and defaulted whenever it could beat never borrowing by a third, and borrowing in the final round was free. With the 25% cap, the 1.25× default cost and the final-round settlement, in all five trading test worlds serial default loses 1.5–3% to never borrowing, repaying beats defaulting at par by about 1% of the six-round total, and after a 40% devaluation default beats repaying in full — the lesson of the Phase 6 crisis round (`stress_test_money_balance.py`).

Divergence. No inter-temporal optimization (no Euler equation, no forward-looking default decision); the borrow/service welfare scaling is a mechanical multiplier. Debt is denominated only in the reserve currency. The rate rule is a reduced-form spread, not a default-probability-priced yield.

9. Phase 7 — Institutions and power

Theory. Hegemonic stability (Kindleberger, Krasner): a leader supplies public goods — open markets, a stable system — at private cost, and is tempted to free-ride. Regimes/the WTO (Keohane) lower transaction costs among members.

Engine (`_institutional_friction_delta`, `_global_crisis_factor`, `challenge_hegemon`). All institutional effects act on *trade friction* or welfare. The friction added to a trade leg, on top of the Phase-5 base and any Phase-6 default friction, is

$$\Delta_{\text{inst}} = \underbrace{\big(-0.05 \text{ if provide else } +0.07\big)}_{\text{hegemon public good}} \quad \underbrace{-\,0.02}_{\substack{\text{WTO dividend, if both members} \\ \text{\& neither defected} \\ (\times 0.5 \text{ if debt-banned})}},$$

then the total fx is clamped to $[0, 1]$. The hegemon pays $U \times= 0.96$ when it provides. A pending **global crisis** of severity ς scales every country's welfare by

$$1 - \varsigma \cdot \big(0.5 \text{ if hegemon provides}\big) \cdot \big(1 - 0.5\,\text{wto_share}\big),$$

so leadership and broad membership both soften it. A **hegemonic challenge** is decided by the same rule as a reserve-currency challenge (Phase 5): by default the coalition must be a majority of the countries, or under `"welfare"` its cumulative welfare must exceed that of all non-coalition countries; hegemony and the reserve-currency role then transfer. Side payments are goods transfers applied before welfare. They land once the trades have settled, and a trade still short of goods is then retried, so goods received as a side payment can be passed on in a trade the same round (and a side payment can hand on an import).

Divergence. A stylized public-goods/coordination model, not a game-theoretic equilibrium. The challenge rule is a head count (or, by choice, a raw weight comparison), not a bargaining solution; "defection" is simply applying a tariff above a binding. The hegemony parameters are tuned (see

the simulation notes) so that provision is collectively beneficial in both WTO-none and WTO-all settings while the hegemon is still individually tempted to free-ride.

All formulas above are exactly what the code computes; the `stress_test_phase*.py` *suites assert these numbers. To change a behavior, edit the constant at the top of* `engine.py` *and rerun the tests.*